

Code-Layout, Readability and Reusability

Date	7 July, 2026
Team ID	
Project Name	Machine Learning-Based Flood Prediction and Early Warning System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the Machine Learning-Based Flood Prediction and Early Warning System in terms of code organization, readability, maintainability, and reusability. The project is developed using Python, Flask, Scikit-learn, XGBoost, HTML, CSS, and JavaScript. Proper coding standards, modular programming practices, and meaningful documentation ensure that the application is easy to understand, maintain, and extend for future improvements.Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform indentation and spacing maintained throughout the Python code.	Yes	Improves readability and debugging.
2	Proper File Structure	Files are organized into templates, static, model, dataset, and application folders.	Yes	Follows Flask project structure
3	Meaningful Variable Names	Variables such as annual_rainfall , cloud_visibility , seasonal_rainfall , flood_prediction clearly indicate their purpose.	Yes	Easy to understand
4	Function / Method Names	Functions like predict(), load_model(), and preprocess_data() are descriptively named.	Yes	Enhances maintainability.
5	Code comments	Important sections contain comments explaining preprocessing, model loading, and prediction logic.	Yes	Helps future developers.
6	Modular Design	Data preprocessing, model training, evaluation, and web application are separated into different modules.	Yes	Highly reusable.
7	No Redundant Code	Duplicate code has been minimized by using reusable functions.	Yes	Cleaner implementation.

8	Error Handling	Invalid user inputs and prediction errors are handled using exception handling.	Yes	Prevents application crashes.
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Used	Reusability Level (High / Medium / Low)
1	Flask Web Application	Flask, HTML, CSS	User Interface	High
2	Data Preprocessing Module	Python, Pandas, NumPy	Weather Data Preparation	High
3	Machine Learning Model	Decision Tree, Random Forest, KNN, XGBoost	Flood Prediction System	High

Overall Code Quality Assessment:

Aspect	Rating(1-5)	Comments
Code Layout & Structure	5	Well-organized project directory following Flask architecture.
Readability	5	Meaningful variable names, proper formatting, and comments improve readability.
Reusability	5	Modular functions and reusable ML components allow future enhancements.
Documentation / Comments	5	Adequate comments explain important sections of the code.
Overall Score	5/5	Excellent code quality with proper structure, readability, and maintainability.